CELESTIAL MECHANICS × AI · 2026

From the next token to the next theorem

How a model works — and how I use it in practice

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What the model sees: ten integers

EN · ONE SENTENCE

Celestial mechanics is the poetry of orbits.

Cel	estial	_mechanics	_is	_the	_poetry	_of	_or	bits	.
69126	90010	52347	382	290	38153	328	503	24043	13

46 characters → **10** tokens → 10 integers · _ marks a leading space

CHINESE · FOUNDATIONS OF CELESTIAL MECHANICS

本书系统地介绍了天体力学

本	书	系统	地	介绍	了	天	体	力	学
4087	20121	31936	5557	37887	4531	867	7074	11343	8590

12 characters → **10** tokens → 10 integers · two-character words often map to one token

“This book systematically introduces celestial mechanics.”

01 Where do these integers come from?

02 How does it predict the next one?

03 How do we put it to work?

RL: actions, outcomes and feedback

Act in an environment; use outcome feedback to improve action choices.

Observe environment → Take action → Outcome → Feedback / reward → Improve policy ↺

ENVIRONMENT

Produce an outcome

Actions affect the environment; new observations return.

FEEDBACK

Evaluate the result

Reward expresses progress or success.

POLICY

Improve action choices

Use experience to seek higher long-run return.

> RL uses **outcome feedback** to learn **how to choose actions**.

Reference: Sutton & Barto, Reinforcement Learning: An Introduction.

Math and code share a feedback advantage

01 · CLOSED ENVIRONMENT

Axioms · Runtime

Math uses definitions, axioms and inference rules; code uses a language spec and runtime. The system is self-contained.

02 · TRACEABLE ERRORS

Locate the step

Proofs check inference step by step; tests, traces and line numbers locate code errors.

03 · CLEAN DATA

Parse and filter

Formal objects and code are structured, enabling bulk generation and verification.

Two levers: better policies + more compute for candidate generation and verification.

Closed environment + localizable errors + clean data → **cheap feedback, large-scale trial and error**.

Closure is relative: code still depends on inputs and libraries; modeling still needs formalization; data still needs curation.

Summary: environment, feedback, updates

Well-defined tasks let a model predict, receive checks and iterate in an environment.

Problem + environment → Next-token prediction → Check + trace → Define loss → Backpropagate ↺

01 · ENVIRONMENT

Clear rules & goal

The model can act, try and produce outputs.

02 · FEEDBACK

Evaluate & trace

Check outcomes and locate failing steps.

03 · UPDATE

Loss-driven iteration

Backpropagate, update parameters, improve.

> The environment supplies feedback; **loss turns it into parameter updates**.

The GRPO objective

DeepSeekMath: group-relative advantage with clipping and a KL constraint.

$$J_{\mathrm{GRPO}}\left(\theta\right) = E_{q \sim P(Q),\, \{o_i\}_{i=1}^{G} \sim \pi_{\theta_{\mathrm{old}}}} \left[\frac{1}{G} \sum_{i=1}^{G} \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} L_{i,t} \right]$$

PER-TOKEN TERM

$$L_{i,t} = \min\left[\rho_{i,t}\left(\theta\right) \hat{A}_{i,t},\, \mathrm{clip}\left(\rho_{i,t}\left(\theta\right), 1-\varepsilon, 1+\varepsilon \right) \hat{A}_{i,t} \right] - \beta D_{\mathrm{KL}}\left[\pi_\theta \| \pi_{\mathrm{ref}} \right]$$

PROBABILITY RATIO

$$\rho_{i,t}\left(\theta\right) = \frac{\pi_\theta\left(o_{i,t} \mid q, o_{i,<t}\right)}{\pi_{\theta_{\mathrm{old}}}\left(o_{i,t} \mid q, o_{i,<t}\right)}$$

GROUP-RELATIVE ADVANTAGE

$$\hat{A}_{i,t} = \frac{r_i - \mathrm{mean}\left(r_1, \ldots, r_G\right)}{\mathrm{std}\left(r_1, \ldots, r_G\right)}$$

Raise high-advantage responses while **clip and KL** limit policy drift.

Source: Shao et al., *DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models*, arXiv:2402.03300, 2024 · GRPO objective; implementation details vary.

GRPO: compare a group of answers

No critic; use relative performance within a group.

One question → Sample a group → Score each → Relative advantage ↻

GROUP NORMALIZATION

$$A_i = \frac{r_i - \text{mean}(r)}{\text{std}(r)}$$

Above the group mean → positive; below → negative.

NO CRITIC

The group mean is the baseline, so no separate value model is needed.

STABLE UPDATES

Clipped probability ratios and a KL penalty limit policy drift.

> GRPO learns from **relative quality**, not absolute scores.

REFLEX: search engine in the RL loop

E-commerce query expansion: generate → retrieve → reward result quality.

Query + top-k products → LLM expands → Search engine → Union of results → Reward ↺

REWARD · NO HUMAN LABELS

Relevance Rel(Y)/Rel(q) + coverage Ret(Y)/Ret(q); format violations score 0.

TRAINING

SFT warm start on 10k → GRPO on 100k real queries; 8×H200.

QUERIES WITH POSITIVE GAIN · %

SETTING	ΔCOV	ΔREL
SFT ONLY	12.8	8.9
GRPO ONLY	57.5	25.2
WARM START + GRPO	72.8	32.4

10,000 queries · 5 samples · LLaMA 3.2-3B

> GRPO turns **better retrieval** into an optimizable reward.

Zhang, Liu, et al., REFLEX, WSDM 2026 · ablation on this dataset and backbone.

When code outruns human review

Claude Code's review problem: code scales faster than human attention.

SCALE PRESSURE · ANTHROPIC MONOREPO

2,500 → 30,000+

Changes landing per day · May → Sep 2026

Claude Code has a separate repository.

PRODUCTION

Agents write more code, faster.

BOTTLENECK

Human review time stays limited.

CONSEQUENCE

Line-by-line review cannot keep up.

Goal: reduce **review + repair time**, with reverts and incidents as guardrails.

The next gain comes from **redesigning review**.

Source: Osmani, Lunt & Wang · 15 Oct 2026 · How the Claude Code team does code review · Team-reported practice.

From the next token to the next theorem

`token` → `probability` → `candidate` → `verification` → **`theorem`**

Probability gives breadth

Verification gives correctness

People set the direction

> The next theorem still begins with the **next token**.